

Little Sub and Traveling

Tags:

Time Limit: 1 s Memory Limit: 256 MB
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Codes

AI Code Tips

Description

Little Sub lives in a country with n cities $A[0..n - 1]$. He loves traveling very much.

It is Feb.14th, the Valentine Day. Little Sub decides to travel around the country with his girl friend.

Little Sub lives in $A[0]$ and he wants to visit each city exactly once and go back to $A[0]$ eventually. However, Little Sub has made a rule for his journey: Suppose he is in city $A[i]$, the next city he is about to visit must be either $A[(i \oplus 2) \% n]$ or $A[(i \oplus 2 + 1) \% n]$.

Please help him find a possible route. If there are multiple answers, please give the one with the maximum alphabet order.

We say route A is larger than route B in the alphabet order when there exists i that $A[i] > B[i]$ & $A[1..i - 1] = B[1..i - 1]$ meets.

Input

The first line contains a positive integers n ($2 \leq n \leq 10000$).

Output

If there exists a legal route, please output $n + 1$ integer separated by spaces in a single line, indicating the route.

If there is no possible route, please output -1 instead.

Samples

input	Copy
4	
output	Copy
0 1 3 2 0	
input	Copy
3	
output	Copy
-1	

Author

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Submit

Codes

HZNUOJ is based on HUSTOJ

--- Maintainer List ---

Server Time: 2025-8-26 16:07:58